Matthew Zhang

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Education

The University of Edinburgh — Edinburgh, UK
Master of Informatics (MInf) in Computer Science — *Expected May 2027*

- Current grade: First Class
- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Security, Databases, Machine Learning, Natural Language Processing, Compiling Techniques

Experience

NASA — New York, US
Software Engineering Intern — *June – August 2021*

- Designed a Python data pipeline to process and visualize terabytes of earthquake precursor signals in 3D, enabling researchers to identify anomalies in seismic patterns more efficiently.
- Integrated the NASA WorldWind API into the analysis tool, giving scientists an interactive, globe-based interface for exploring geospatial data.

Edinburgh University — Edinburgh, UK
Tutor — *September 2023 – 2024*

- Tutored Introduction to Computation (Haskell) to cohorts of 10+ first-year computer science students, clarifying core concepts in recursion and proof-based reasoning.
- Led weekly interactive tutorials and provided detailed code-review feedback to improve student algorithmic problem-solving.

Projects

Robotic Pool-Playing Interface *(Node.js, WebSockets, Python/OpenCV, HTML5 Canvas)*

- Engineered a real-time, full-stack teleoperation interface, allowing users to control a physical pool-playing robot via an interactive "drag-and-play" digital twin.
- Architected a low-latency WebSocket communication pipeline integrated with a custom Python computer vision system for dynamic ball tracking.
- Authored a custom 2D physics engine to handle realistic collisions, shot trajectories, and automated turn-taking, enabling seamless human–robot gameplay.

Voice-Guided Video Segmentation for Microscopy *(Python, PyTorch, OpenCV, Whisper, MicroSAM, SAM 2)*

- Built a human-in-the-loop ML pipeline combining voice prompts (Whisper) and point-click interaction (MicroSAM, SAM 2) for expert-guided microscopy video analysis.
- Architected an end-to-end inference engine integrating real-time speech-to-text, intent parsing, temporal propagation, and automatic correction workflows.
- Evaluated model performance across 19 complex video datasets, improving temporal consistency metrics by over 7x compared to baseline methods.
- Developed a reproducible experimentation framework supporting automated CI/CD-style reporting, outputting publication-ready data tables and figures.

Drone Delivery Pathfinding & Order Validation System *(Java, Spring Boot)*

- Engineered an A* pathfinding algorithm to compute optimal drone delivery routes around no-fly zones, reducing runtime from 2s to ˜200ms by eliminating redundant node expansion with a visited-set (closedSet) tracking approach.
- Designed a bounding-box pre-filtering step for geometric collision checks, cutting no-fly-zone intersection testing from O(mn) to O(kn) complexity, and refactored an initially monolithic controller into modular, testable components (validation, pathfinding, geometry).
- Built a REST API (Spring Boot) with comprehensive JUnit test coverage across endpoint, validation, and geometry logic, using MockMvc and mocked external dependencies to ensure isolated, reliable test runs.

Technical Skills

Languages: Python, Java, JavaScript, SQL, C, Haskell
Frameworks & Libraries: React.js, Node.js, Express, Pandas, NumPy, TensorFlow, PyTorch
Tools and Systems: Git/GitHub, Docker, Linux, ROS, REST APIs, WebSockets